

The Bullet Cluster Under EFC: Confronting Entropy-Gradient Gravity with JWST-Era Mass Reconstructions

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Abstract. The Bullet Cluster (1E 0657–56) has long served as a benchmark system for dark matter physics. Three JWST-era studies published in 2025–2026 now provide the highest-resolution mass reconstruction to date (Cha et al. 2025; 146 strong-lensing constraints, 398 sources arcmin^{−2}), the first spectroscopically anchored parametric lens model (Rihtaršič et al. 2026; 135 secure multiple images with z_{spec} from 27 systems), and a definitive mass-ratio measurement establishing the system as a $\sim 10:1$ minor merger (Cho et al. 2025; $M_{200c} = 15.1 \times 10^{14} M_{\odot}$ main + $1.5 \times 10^{14} M_{\odot}$ sub). We confront these results with the Energy-Flow Cosmology (EFC) framework across four axes: (i) reanalysis of the cluster merger geometry structural constraint under the revised mass ratio; (ii) updated predictions for the pre-registered directional lensing residual estimator A_{sig} ; (iii) reinterpretation of the tightened SIDM cross-section bound $\sigma/m < 0.5 \text{ cm}^2 \text{ g}^{-1}$ within EFC’s no-particle ontology; (iv) a first qualitative confrontation of EFC’s regime-transition prediction ($\mu \approx 1.1$, $\Sigma \approx 1.2$ at cluster scales) with the new mass profiles. We find that all three studies are consistent with EFC’s structural predictions and that the 10:1 mass ratio *simplifies* the geometric exclusion argument. The JWST spectroscopic dataset now provides sufficient precision to execute the A_{sig} shock-front asymmetry test, which remains the only pre-registered observable capable of directly discriminating between EFC and Λ CDM at this scale. No new free parameters are introduced.

1 Introduction

The Bullet Cluster occupies a unique position in observational cosmology: it is the system most frequently cited as definitive evidence for particle dark matter, because the gravitational lensing mass peaks are spatially offset from the dominant baryonic component (the X-ray-emitting intracluster medium, ICM) after a high-velocity ($\sim 4500 \text{ km s}^{-1}$) cluster merger.

In the Λ CDM interpretation, two collisionless dark matter haloes passed through each other while the collisional gas experienced ram-pressure stripping. Any modified gravity theory must explain the same mass–gas offset without invoking collisionless particles.

Energy-Flow Cosmology (EFC) addresses this through three mechanisms:

1. **Density Saturation** $\Theta(\rho)$: a smooth function that suppresses EFC corrections in high-density environments ($\rho \gg \rho_{\text{crit}}$), recovering GR in Solar System and stellar interior regimes.
2. **Regime-dependent gravitational modification**: the EFC Relativistic Action produces $\mu < 1$ (growth suppression) at cosmological linear scales and $\mu > 1$ (gravitational enhancement) at non-linear galactic/cluster scales, from a single variational action.
3. **Entropy-gradient lensing**: the effective gravitational potential traced by lensing is coupled to the entropy-flow scalar field φ , producing anisotropic lensing responses near thermodynamic discontinuities (shock fronts).

This paper does not introduce new parameters, simulations, or data reductions. It is an analytical confrontation of existing EFC predictions—all documented in the EFC Validation Ledger (v3.11)—with three new observational results from the JWST era.

Ledger context (v3.10–v3.11): The WP4 BOSS Transfer Validation (v3.10, DOI: 31954125) demonstrates that EFC parameters calibrated on DESI DR2 ($z_{L1L2} = 1.01$, $\alpha_{L2} = 0.045$) transfer frozen to BOSS DR12 with $\Delta\chi^2 = -7.77$. The Multi-Epoch $f\sigma_8$ Growth Test (v3.11, DOI: 31955871, EFC-VAL-2026-003) yields a *null result* for the perturbation sector: $B = 0.078$, $\mu_0 = 0.922$, $\Delta\chi^2 = 0.10$ (0.06σ). The pilot 1.7σ preference for $\alpha_{L2} > 0$ weakens to $< 0.1\sigma$ once regime-dependent $\mu(a)$ correctly separates BAO geometry from RSD growth. Together, these establish that EFC is *consistent* with growth data but does not yet show a perturbation-sector detection—the background sector carries the signal (WP4), not the growth ODE.

2 New Observational Inputs

2.1 Cha et al. (2025): High-resolution mass map

Cha et al. (2025, ApJ 987, L15; arXiv:2503.21870) present the highest-resolution mass reconstruction of the Bullet Cluster to date, combining JWST NIRCам strong lensing (146 constraints from 37 systems) with weak lensing at a source density of 398 arcmin^{-2} . Key results:

- The main cluster’s mass distribution is highly elongated (NW–SE), consisting of at least three subclumps aligned with the brightest cluster galaxies (BCGs).
- The subcluster is compact, with a single dominant mass peak.
- A possible mass and intracluster light (ICL) trail extends from the subcluster eastward toward the main cluster.
- ICL traces dark matter with a modified Hausdorff distance of $19.80 \pm 12.46 \text{ kpc}$.
- The tightest constraint on dark matter self-interaction to date: $\sigma/m < 0.5 \text{ cm}^2 \text{ g}^{-1}$.

2.2 Rihtaršič et al. (2026): Spectroscopic lens model

Rihtaršič et al. (2026, arXiv:2601.22245) present an updated parametric lens model combining JWST NIRCам imaging and NIRSpectroscopy. The model is constrained by 135 secure multiple images from 27 background galaxies with spectroscopic redshifts spanning $z = 0.9\text{--}6.7$ (previously only 6 systems had spectroscopic z). The main cluster is described by a complex, elongated double-peaked PIEMD distribution; the subcluster by a single large-scale halo with $\sim 4 \text{ kpc}$ offset from the BCG.

2.3 Cho et al. (2025): Minor merger confirmation

Cho et al. (2025, arXiv:2512.03150) combine JWST/NIRCам strong lensing with DECam wide-field weak lensing to obtain the first robust virial masses for three individual components:

$$M_{200c}^{\text{main}} = 15.11_{-2.10}^{+2.48} \times 10^{14} M_{\odot}, \quad M_{200c}^{\text{sub}} = 1.49_{-0.25}^{+0.32} \times 10^{14} M_{\odot}. \quad (1)$$

The mass ratio is $\sim 10:1$, establishing the Bullet Cluster as a **minor merger**—not the $\sim 3:1$ or $6:1$ ratio assumed in many earlier simulations.

3 Confrontation with EFC Predictions

3.1 Cluster merger geometry (updated)

The structural exclusion test (Magnusson 2026, DOI: 31173850) used cluster lensing mass–gas offsets as an L2 geometric constraint to rule out certain EFC operator classes. The test established that local entropy-gradient couplings producing isotropic modifications are disfavoured, and that admissible operators must possess component sensitivity and geometric smoothing.

Under the revised 10:1 mass ratio (Cho et al. 2025), this constraint is *strengthened*, not weakened:

1. A minor merger produces a more asymmetric gravitational potential, which amplifies the distinction between isotropic and anisotropic lensing responses.
2. The subcluster mass ($1.5 \times 10^{14} M_\odot$) places it firmly in the regime where $\Theta(\rho)$ provides partial—but not complete—screening ($\rho_{\text{cluster}} \sim 10^{-2}\text{--}10^{-1} \rho_{\text{crit}}^\Theta$).
3. The three-subclump structure of the main cluster (Cha et al.) introduces additional geometric constraints: any entropy-gradient coupling must produce lensing peaks aligned with *all three* subclumps, not just a single halo.

Result: The existing structural exclusion test survives and is tightened under the new mass ratio. No operator class previously excluded is restored.

3.2 A_{sig} estimator: method and baseline result

The pre-registered directional lensing residual estimator A_{sig} (methodology locked 2026; documented in EFC Validation Ledger v3.11) measures whether lensing convergence residuals $\delta\kappa = \kappa_{\text{obs}} - \kappa_{\text{density}}$ show shock-locked asymmetry after density-based mass model subtraction.

EFC prediction: Non-zero A_{sig} aligned with the Chandra shock front, produced by entropy-gradient coupling between the scalar field φ and the metric response Σ .

Λ CDM prediction: $A_{\text{sig}} = 0$ (isotropic residuals after mass model subtraction).

3.2.1 PIEMD baseline model

We construct a 2D convergence map $\kappa(x, y)$ on a 512×512 grid covering 1800×1200 kpc using analytical Pseudo-Isothermal Elliptical Mass Distributions (PIEMDs; Kassiola & Kovner 1993; Kneib et al. 1996). Each halo has seven parameters: centre (x_0, y_0) , velocity dispersion σ_0 , core radius r_{core} , truncation radius r_{cut} , ellipticity $\varepsilon = (a - b)/(a + b)$, and position angle θ . The circular PIEMD projected surface density is:

$$\Sigma(R) = \frac{\sigma_0^2}{2G} \cdot \frac{r_{\text{cut}}}{r_{\text{cut}} - r_{\text{core}}} \left[\frac{1}{\sqrt{r_{\text{core}}^2 + R^2}} - \frac{1}{\sqrt{r_{\text{cut}}^2 + R^2}} \right], \quad (2)$$

with elliptical extension via $R_\varepsilon^2 = x'^2/(1 - \varepsilon)^2 + y'^2/(1 + \varepsilon)^2$, where (x', y') are coordinates rotated by the position angle θ . The convergence is $\kappa = \Sigma/\Sigma_{\text{cr}}$ with $\Sigma_{\text{cr}} = 2.84 \times 10^9 M_\odot \text{ kpc}^{-2}$ for $z_l = 0.296$, $z_s = 1.0$.

We model the system with four large-scale halos motivated by Rihtaršič et al. (2026): Main-NW, Main-SE (the double-peaked main cluster), Subcluster (the bullet), and a Group-NE extension. Placeholder parameters are estimated from Cho et al. (2025) virial masses and Cha et al. (2025) reported offsets; these will be replaced with the exact best-fit values from Rihtaršič et al. Table 4 in the final version.

The resulting κ map produces peak values of $\kappa \approx 0.62$ (Main-NW), 0.49 (Main-SE), 0.49 (Subcluster), and 0.16 (Group-NE), consistent with published convergence values from Cha et al. (2025).

3.2.2 Asig operator (2D)

We define the shock front as a line at $x_{\text{shock}} = 480$ kpc with normal $\hat{n} = (-1, 0)$ (pointing westward, toward the main cluster), following Markevitch (2006).

Two rectangular strips are defined parallel to the shock front, each 200 kpc wide and 400 kpc long:

- **Front** (pre-shock, toward main cluster): $0 < d_{\perp} < 200$ kpc
- **Back** (post-shock, behind bullet): $-200 < d_{\perp} < 0$ kpc

where $d_{\perp}$ is the signed perpendicular distance from the shock line. The 2D A_{sig} estimator is:

$$A_{\text{sig}} = \langle \kappa \rangle_{\text{front}} - \langle \kappa \rangle_{\text{back}} . \quad (3)$$

3.2.3 Baseline result

Table 1: A_{sig} 2D baseline (PIEMD model, placeholder parameters).

Quantity	Value
$\langle \kappa \rangle_{\text{front}}$	0.1824
$\langle \kappa \rangle_{\text{back}}$	0.1831
A_{sig} (baseline)	-6.9×10^{-4}
Bootstrap p -value (5000 random rotations)	0.98
$A_{\text{sig}}/\sigma_{\text{null}}$	-0.01σ
Sensitivity ($\pm 20\%$ in $\sigma_0^{\text{Main-NW}}$)	$\Delta A_{\text{sig}} \approx \pm 0.01$

The parametric gravitational model produces **no directional asymmetry** at the shock front ($A_{\text{sig}} \approx 0$, $p = 0.98$). This result establishes the null hypothesis: in the full test, any $A_{\text{sig}} \neq 0$ in the residual map $\delta\kappa = \kappa_{\text{obs}} - \kappa_{\text{PIEMD}}$ cannot be attributed to halo geometry within this parametric model class.

Caveat: This is a model-level diagnostic with approximate parameters. The result shows that PIEMD halo geometry does not generate shock-locked asymmetry, but the residual $\delta\kappa$ cannot be computed without the Cha et al. (2025) free-form κ -map. The baseline null validates the test operator, not the physical hypothesis.

The complete pipeline (PIEMD projection, Asig operator, bootstrap, and sensitivity analysis) is provided as open-source Python code in the supplementary materials.

3.3 SIDM constraint: $\sigma/m < 0.5 \text{ cm}^2 \text{ g}^{-1}$

Cha et al. (2025) derive the tightest upper bound on the dark matter self-interaction cross section from the observed offset between the subcluster BCG position and the lensing mass peak ($\sim 60 \text{ kpc}$).

In EFC, this result has a different interpretation:

1. EFC does not postulate dark matter particles; the lensing mass is produced by entropy-gradient gravitational coupling ($\Theta(\rho)$ -modulated Σ).
2. The “collisionless” behaviour of the lensing mass—passing through the opposing cluster while gas is stripped—is *expected* in EFC because the entropy-gradient field responds to the total gravitational potential $\nabla\Phi$, which is dominated by the galaxy distribution (collisionless stellar component), not the gas.
3. Progressive tightening of SIDM bounds toward zero is *consistent* with a scenario where no DM particle exists. However, it does not *prove* this—collisionless CDM ($\sigma/m = 0$) is equally consistent.

Discriminator: The SIDM bound alone does not distinguish EFC from ΛCDM . Both predict $\sigma/m \approx 0$. The discriminating observable is A_{sig} : EFC predicts non-zero shock-locked asymmetry, while ΛCDM with collisionless CDM predicts isotropic residuals.

3.4 Regime-transition prediction P1

The Regime Transition Test (Magnusson 2026, DOI: 31941543) predicts that at cluster scales (non-linear, L2 regime), the EFC action produces:

$$\mu_{\text{cluster}} \approx 1.1, \quad \Sigma_{\text{cluster}} \approx 1.2, \quad (4)$$

where $\mu > 1$ indicates gravitational *enhancement* (opposite to the cosmological linear regime where $\mu < 1$). This is the regime transition driven by the stiffness response $R \propto k^{-4}$.

The Cho et al. virial masses can be used for a qualitative consistency check: if EFC enhances the effective gravitational mass by $\sim 10\%$ at cluster scales (via $\mu \approx 1.1$), then the *baryonic* mass required to produce the observed lensing signal is reduced by the same factor. For the main cluster:

$$M_{\text{baryonic}}^{\text{EFC}} \approx \frac{M_{200c}^{\text{lens}}}{\mu_{\text{cluster}}} = \frac{15.1 \times 10^{14}}{1.1} \approx 13.7 \times 10^{14} M_{\odot}. \quad (5)$$

This remains much larger than the observed baryonic mass ($\sim 15\%$ of M_{200c} is gas + stars in typical clusters, i.e. $\sim 2.3 \times 10^{14} M_{\odot}$). The $\sim 10\%$ μ -enhancement is therefore **insufficient** to account for the full mass discrepancy via gravity modification alone at this scale.

Honest assessment: EFC does not claim to eliminate the cluster-scale mass discrepancy through the perturbation-sector μ alone. The Regime Response Surface $R(k, S, \rho)$ predicts that the *combination* of $\mu > 1$, $\Sigma > 1$, and $\Theta(\rho)$ -dependent screening produces an effective mass that differs from ΛCDM —but the full quantitative confrontation requires solving the EFC field equations on cluster-specific density profiles, which is a planned pipeline item (Section 5).

4 ICL–Mass Correlation

Cha et al. (2025) report that the intracluster light traces the lensing mass with a modified Hausdorff distance of 19.80 ± 12.46 kpc. This is notable because ICL consists of stars stripped from galaxies during the merger—a purely collisionless, baryonic component.

In Λ CDM, ICL traces DM because both are collisionless and were co-located before the merger.

In EFC, ICL traces the *effective gravitational mass* because the entropy-gradient coupling $\Theta(\rho)$ responds primarily to the stellar/galaxy distribution (which dominates $|\nabla\Phi|$ at cluster scales outside the ICM core). The tight ICL–mass correlation is therefore equally expected in EFC—the physical mechanism differs (collisionless particles vs. entropy-gradient coupling to collisionless baryons), but the observable prediction is the same.

Potential discriminator: If future studies achieve sub-10 kpc precision on the ICL–mass Hausdorff distance, EFC predicts a *directional* bias in the offset aligned with the entropy gradient (shock front direction), while Λ CDM predicts a statistically isotropic scatter. This is structurally related to the A_{sig} estimator.

5 Discussion

5.1 Summary of confrontation

Table 2: EFC consistency with JWST-era Bullet Cluster results.

Observable	EFC	Λ CDM	Discriminating?
Mass–gas offset geometry	✓	✓	No (both accommodate)
$\sigma/m < 0.5$	✓	✓	No (both predict ≈ 0)
10:1 mass ratio	✓	✓	No (simplifies both)
ICL–mass correlation	✓	✓	Potentially (directionality)
A_{sig} baseline (PIEMD)	≈ 0	≈ 0	No (null validated)
A_{sig} on $\delta\kappa$ residual	Predicted $\neq 0$	Predicted $= 0$	Yes (decisive; pending)

The central conclusion is that none of the three new JWST-era studies falsifies EFC, and none provides unique support for EFC over Λ CDM. The Bullet Cluster remains a *consistency check* for EFC at the cluster scale (L2 regime), not a positive detection site—unless and until the pre-registered A_{sig} test is executed.

Implication of v3.11 null result: The multi-epoch $f\sigma_8$ test (EFC-VAL-2026-003) shows that the perturbation-sector coupling $\mu(a)$ is not detected in current growth data ($B = 0$ within 1σ). This means the Bullet Cluster A_{sig} test—which probes the *lensing* sector (Σ), not the growth sector (μ)—is now even more important: the survival valley ($\mu \approx 0.94$, $\Sigma \approx 1.05$) requires that lensing and growth modifications be decoupled, with $\Sigma > 1$ carrying the observable signal at L2 scales. The Bullet Cluster shock-front is precisely where this decoupling can be tested.

5.2 What has changed

The JWST data transform the *precision* of the Bullet Cluster as a testing ground:

1. The 10:1 mass ratio (Cho et al.) resolves a decades-long ambiguity and strengthens geometric constraints for all gravity theories.
2. The 135 spec- z systems (Rihtaršič et al.) make percent-level κ -reconstruction feasible for the first time.
3. The $\sigma/m < 0.5$ bound (Cha et al.) closes parameter space for SIDM models with $\sigma/m > 0.5$, but does not affect EFC (which predicts $\sigma/m = 0$ by construction).

5.3 Limitations

1. The A_{sig} baseline uses a parametric PIEMD model with placeholder parameters. The result validates the null hypothesis (no geometric asymmetry) but cannot compute the physical residual $\delta\kappa$ without the free-form κ -map from Cha et al. (2025).
2. PIEMD parameters are approximate estimates, not the Rihtaršič et al. best-fit values. The baseline A_{sig} is sensitive to σ_0 at the ± 0.01 level per 20% variation (Table 1).
3. The regime-transition prediction P1 is confronted qualitatively only. No EFC field equations are solved on the new mass profiles.

4. The ICL directionality discriminator is speculative and requires sub-10 kpc precision not yet achieved.
5. EFC’s cluster-scale phenomenology ($\mu > 1$, $\Sigma > 1$) is derived from the Relativistic Action in a perturbative regime. Non-perturbative effects in the shock-front region are not yet computed.

5.4 Next steps

1. **Obtain Cha et al. (2025) κ -map** (free-form reconstruction FITS) and compute $\delta\kappa = \kappa_{\text{obs}} - \kappa_{\text{PIEMD}}$. Apply the A_{sig} pipeline (provided in supplementary materials) to the residual. This is the decisive test.
2. **Replace placeholder parameters** with Rihtaršič et al. (2026) best-fit PIEMD values (Table 4) to refine the baseline.
3. **Cross-check with Abell 2146** (second pre-registered target for A_{sig}) using existing Chandra deep observations and forthcoming JWST data.
4. **Solve EFC field equations** on the Cho et al. virial mass profile to obtain quantitative $\mu(r)$, $\Sigma(r)$ predictions at cluster scale.

6 Reproducibility

All EFC predictions referenced in this paper are documented in the EFC Validation Ledger v3.11, publicly available at:

https://supertedai.github.io/EFC/public/EFC_Validation_Ledger.html

The A_{sig} 2D pipeline (PIEMD analytical projection, multi-halo summation, A_{sig} operator, bootstrap null test, and sensitivity analysis) is provided as open-source Python code (`asig_2d_piemd.py`) in the supplementary materials accompanying this paper.

Citable artifacts: cluster merger geometry (DOI: 10.6084/m9.figshare.31173850), directional lensing residuals (DOI: 10.6084/m9.figshare.31222900), regime transition test (DOI: 10.6084/m9.figshare.31941543), EFC screening model (DOI: 10.6084/m9.figshare.31940469), relativistic action (DOI: 10.6084/m9.figshare.31876324), triangulation test (DOI: 10.6084/m9.figshare.31173850), empirical validation—cluster lensing (DOI: 10.6084/m9.figshare.31190233).

No new parameters are introduced. All parameter values are frozen at their Ledger v3.11 values ($k = 0.415$, $\mu \approx 0.94/1.1$, $\Sigma \approx 1.05/1.2$, $C = 2.32$, $\beta = 0.18$, $\alpha_{L2} = 0.045$, growth-sector $B = 0.078$, $\mu_0 = 0.922$).

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